

VALOQ TERMINAL — COMPLETE INFRASTRUCTURE, VENDOR & GATEWAY MANIFEST

Production: valoq.co.uk | API: api.valoq.co.uk | Stack: Next.js 14, FastAPI, NGINX, Docker, Supabase, Redis, Gemini 1.5

Platform / Tech	Architectural Role	Why We Used It	Actions, Configurations & Modifications Applied
Porkbun porkbun.com	<i>Domain Registrar & Edge DNS</i>	Cost-effective domain registry with flat renewal pricing and free WHOIS privacy.	- Registered domain: valoq.co.uk. A Record: @ → 76.76.21.21 (Vercel Edge). CNAME: www → cname.vercel-dns.com. CNAME: api → valoq-backend.onrender.com.
Vercel vercel.com	<i>Frontend Hosting & Edge CDN</i>	Zero-config serverless Next.js edge platform with automated SSL generation.	- Connected repo: ApexQuantum (directory: apex-frontend). - Injected env: NEXT_PUBLIC_SUPABASE_URL & ANON_KEY. - Bound custom domain: valoq.co.uk with automatic canonical redirect for www.
Render render.com	<i>Backend Compute & API Host</i>	Native Python cloud hosting running FastAPI with custom domain routing.	- Created Web Service: valoq-backend pointing to apex-backend. - Bound custom subdomain: api.valoq.co.uk with edge SSL. - Configured runtime environment variables (Supabase, Redis, Gemini).
NGINX nginx.org	<i>Reverse Proxy & Rate Limiter</i>	Edge gateway managing request routing, Gzip compression, rate limiting, and SSE passthrough.	- Configured nginx/nginx.conf with upstream pools for frontend (:3000) and backend (:8000). - Enforced IP rate limits: 30r/s on general API, 5r/s on AI stream. - Added proxy_buffering off to support uninterrupted Gemini SSE token streaming.
Docker & Compose docker.com	<i>Containerization & Orchestration</i>	Guarantees identical multi-stage runtime environments across local development and production VPS.	- Created multi-stage Node 18 Alpine Dockerfile for Next.js standalone build. - Created slim Python 3.11 Dockerfile for FastAPI. - Authored unified docker-compose.yml tying NGINX, backend, and frontend via bridge network valoq-net.
Supabase supabase.com	<i>Managed PostgreSQL & Cold Storage</i>	PostgreSQL with instant SQL execution, indexed schemas, and relational integrity.	- Provisioned database instance: mzbcvazvagesgymqzhd. - Ran schema migration for watchlist (custom buy target prices, notes). - Migrated equity_prices for historical OHLCV candles with composite indexing. - Migrated equity_financials for normalized corporate statement rows.
Upstash upstash.com	<i>Serverless Hot-Tier Redis</i>	Fast in-memory cache to prevent repeated upstream queries and bypass rate caps.	- Provisioned low-latency TLS Redis database. - Programmed dual-tier caching (Upstash Redis with fallback to in-memory TTLCache). - Set 60s TTL for real-time market quotes and 300s TTL for search queries.
Google AI Studio aistudio.google.com	<i>Gemini 1.5 Flash Reasoning Engine</i>	Fast LLM inference for executive memos, SEC 10-K risk audits, and Bull/Bear models.	- Generated production API key and set GEMINI_API_KEY on Render. - Wired backend route /api/v1/stock/ai-stream using Server-Sent Events (SSE). - Built frontend typewriter stream consumption via Fetch ReadableStream.
Yahoo Finance (`yfinance`) pypi.org/project/yfinance	<i>Market Telemetry Feed</i>	Comprehensive financial data feed covering US (NYSE/NASDAQ), UK (LSE), and Indian (NSE/BSE) equities.	- Integrated real-time quotes, 52-week high/low ranges, calendar earnings dates, and multiples. - Implemented defensive fallback mocking to prevent 500 error cascades during data drops. - Automated 50-day EMA, 200-day SMA, and 14-day RSI calculations.
UptimeRobot uptimerobot.com	<i>Keep-Alive Ping & Uptime Monitor</i>	Keeps free/starter container instances warm by preventing inactivity sleep cycles.	- Configured 5-minute automated HTTPS monitor pinging https://valoq-backend.onrender.com/. - Set up email alerts for downtime or unexpected HTTP status spikes.
GitHub github.com	<i>Version Control & CI/CD Trigger</i>	Central source repository triggering automated deployment pipelines on every commit.	- Structured as a unified monorepo (apex-frontend, apex-backend, nginx). - Git push webhooks trigger Vercel Edge builds and Render deployments automatically on the main branch.